

Training Configuration and Summary:

Sequencing:

- timesteps: 30
- batch_size: 128

Model Hyperparameters:

- min_units: 50
- max_units: 150
- units_step: 50
- dropout_min: 0.2
- dropout_max: 0.3
- dropout_step: 0.1
- activation_functions: ['tanh']
- optimizers: ['adam']

Tuning:

- max_epochs: 20
- hyperband_iterations: 3

BiGRU Multi-Step Training Report

Report Generated: 2025-05-03 20:37:39

Training Start: 2025-04-28_16-04-09

Training End: 2025-04-28_16-09-45

Training Duration: 5.60 minutes

Best Hyperparameters:

Number of Layers: 1

Activation: tanh

Optimizer: adam

Dropout Rate: 0.20

MAE: 2.5921

MSE: 9.0849

RMSE: 3.0141

R2 Score: -0.5018

Multi-Step Forecast:

Forecast Steps: 50

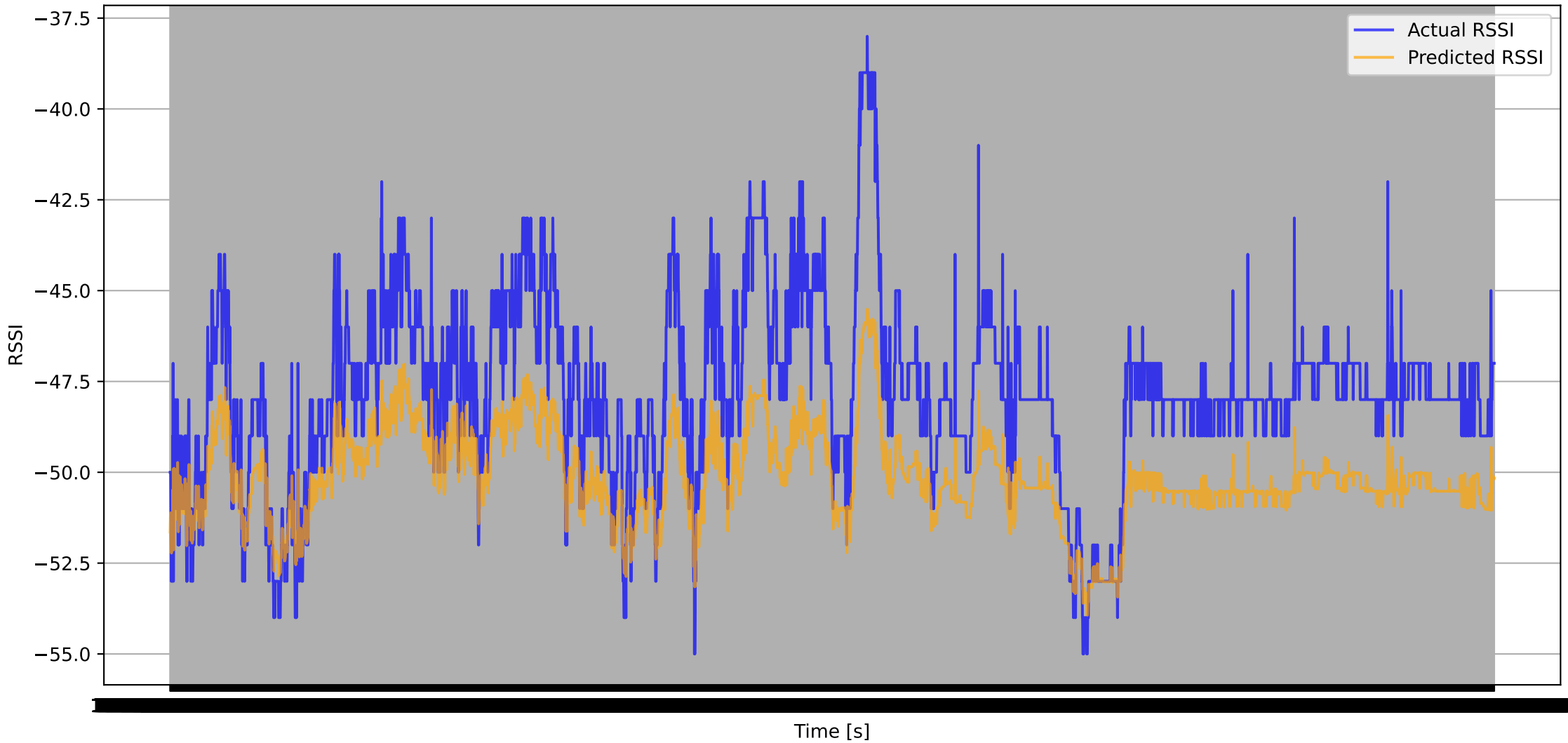
MAE: 0.0826

MSE: 0.0087

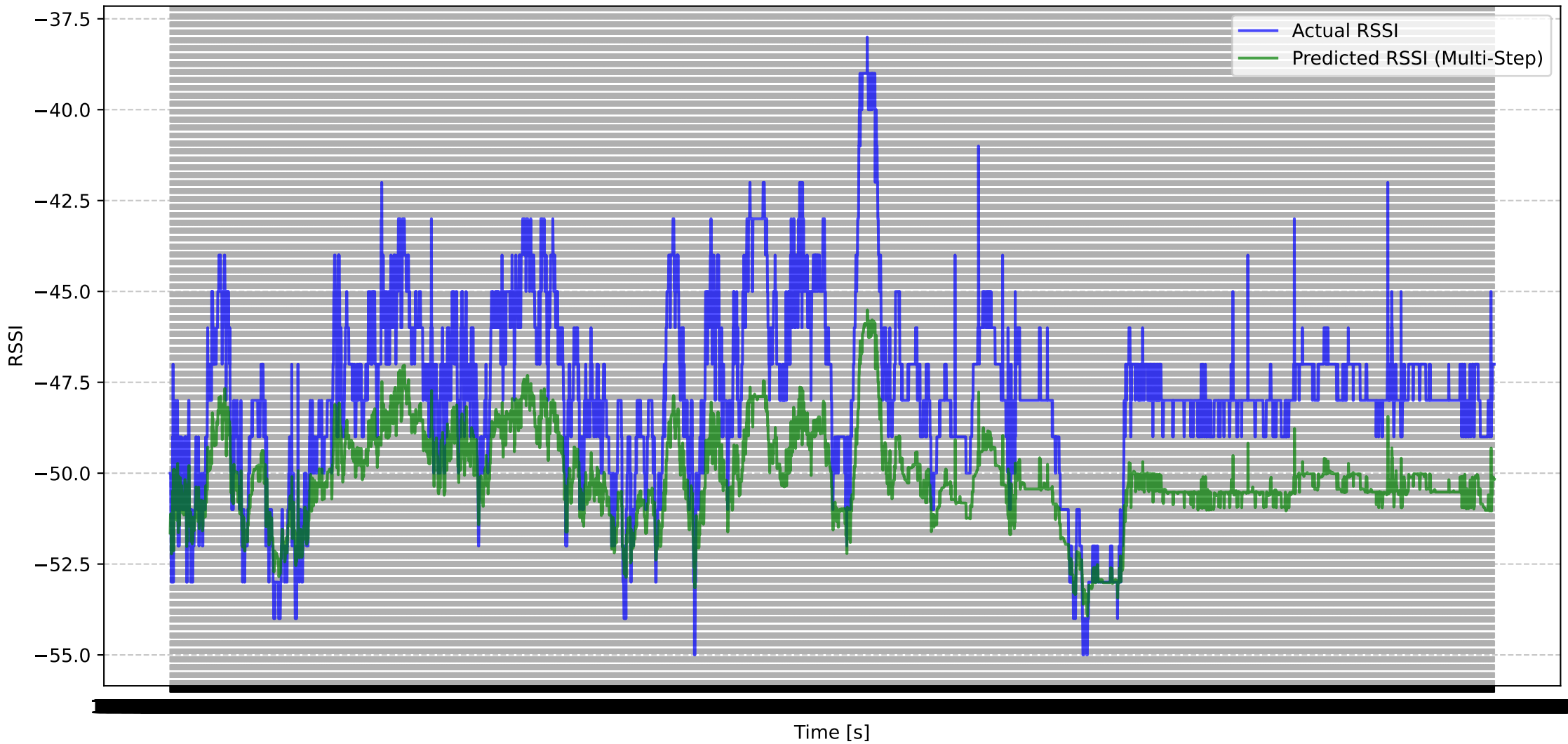
RMSE: 0.0933

R2 Score: -0.1971

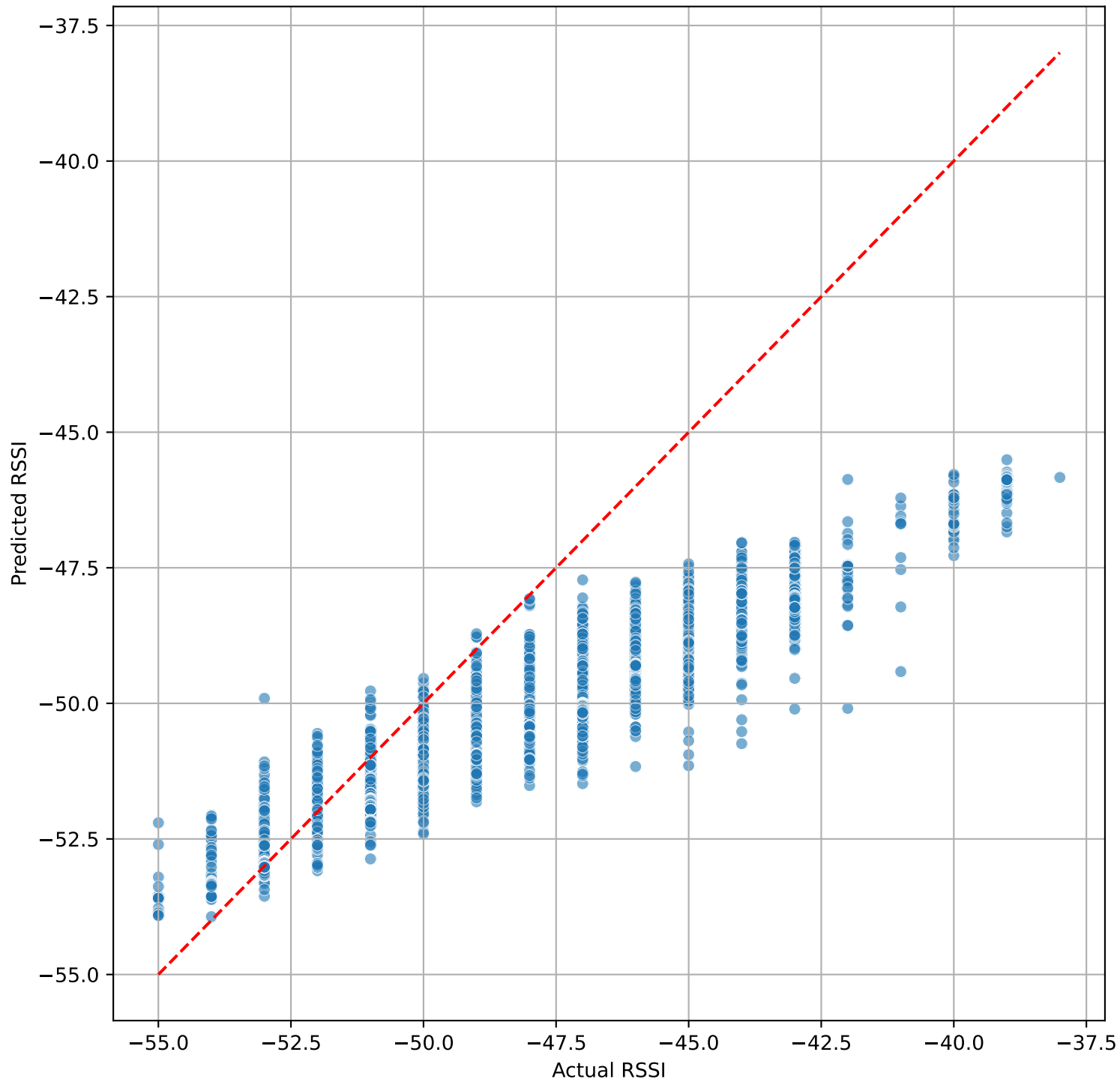
Actual vs Predicted RSSI (First Step)



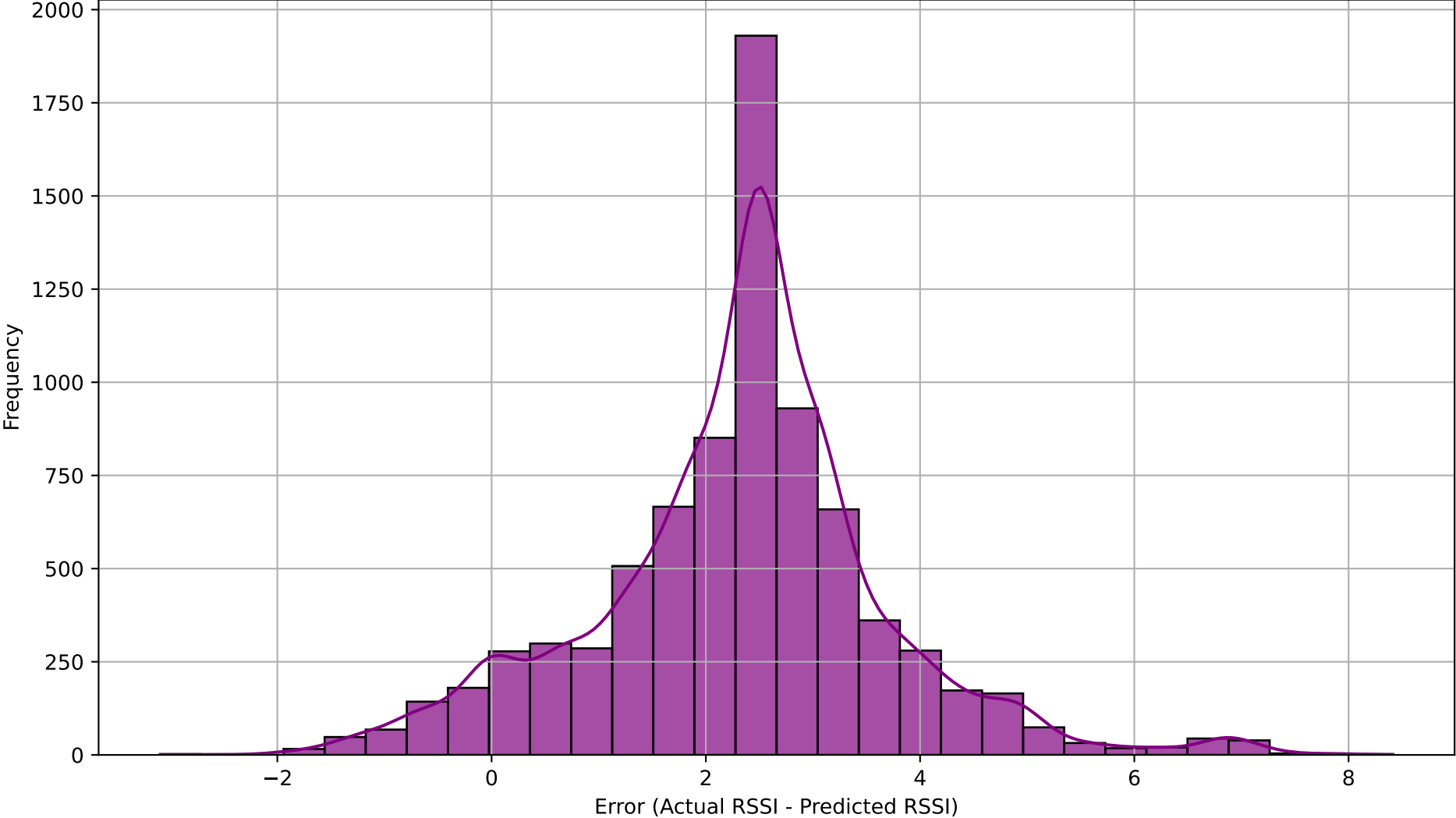
Actual vs Predicted RSSI (First Step vs Multi-Step)



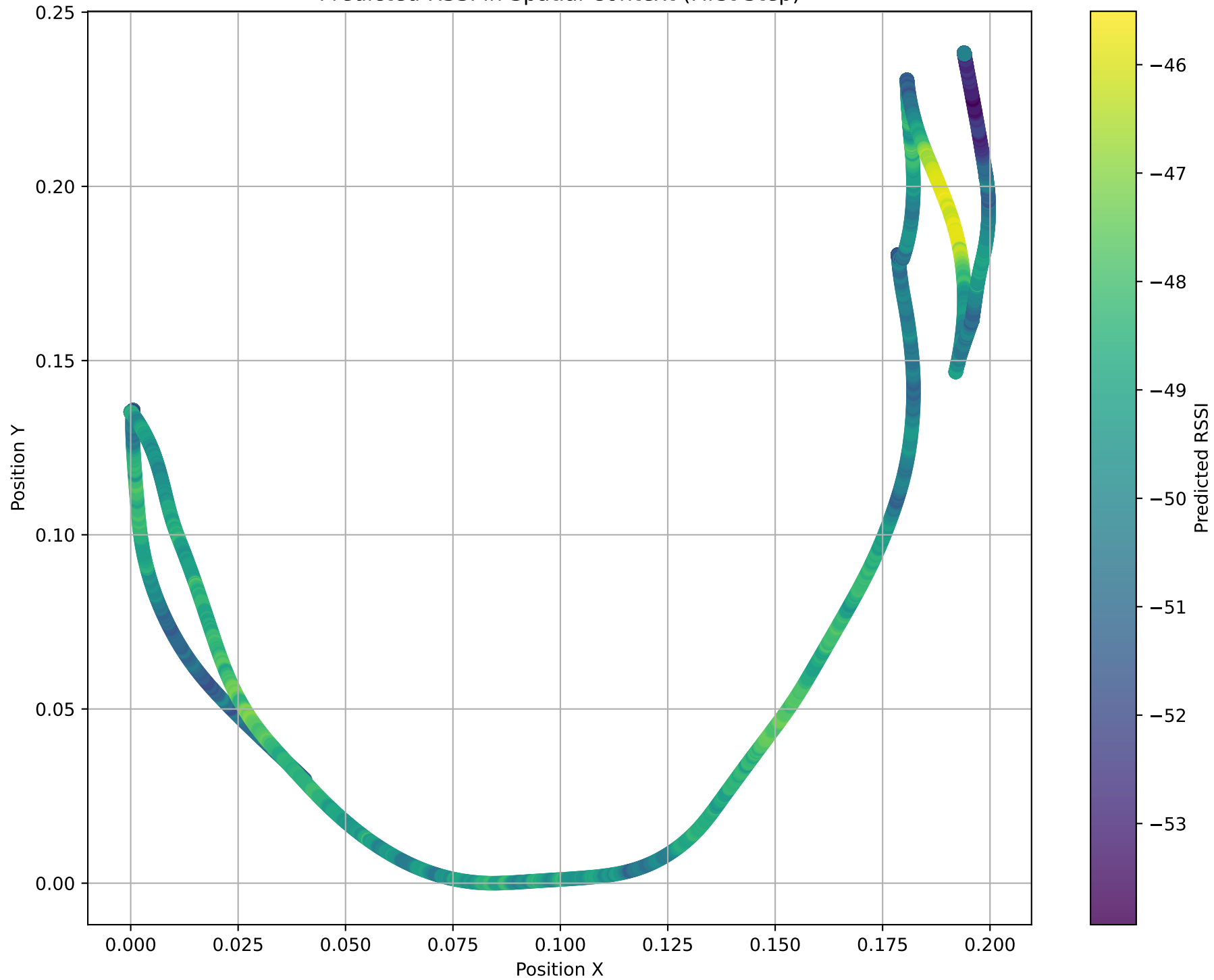
Actual vs Predicted RSSI (First Step)



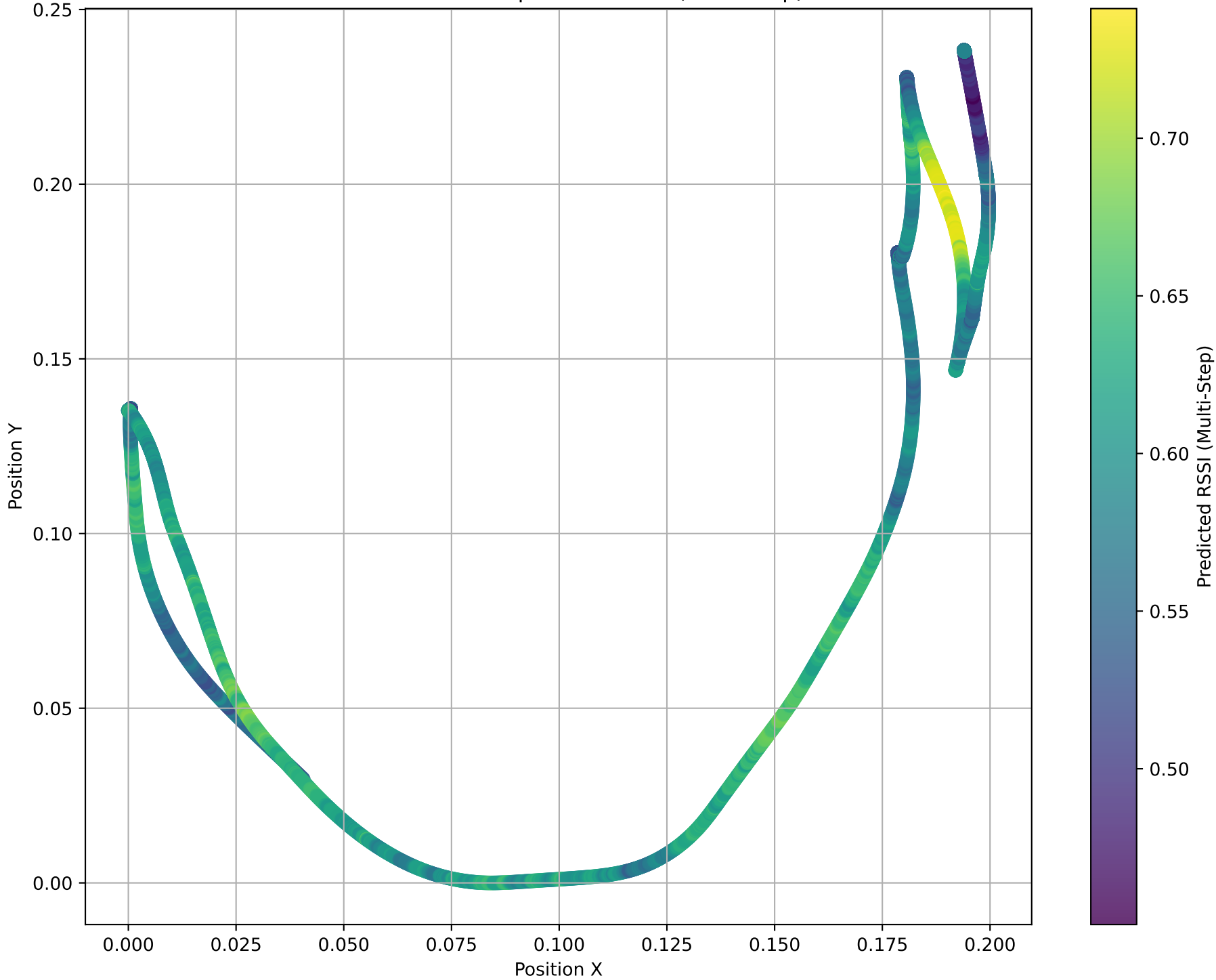
Distribution of Prediction Errors (First Step)



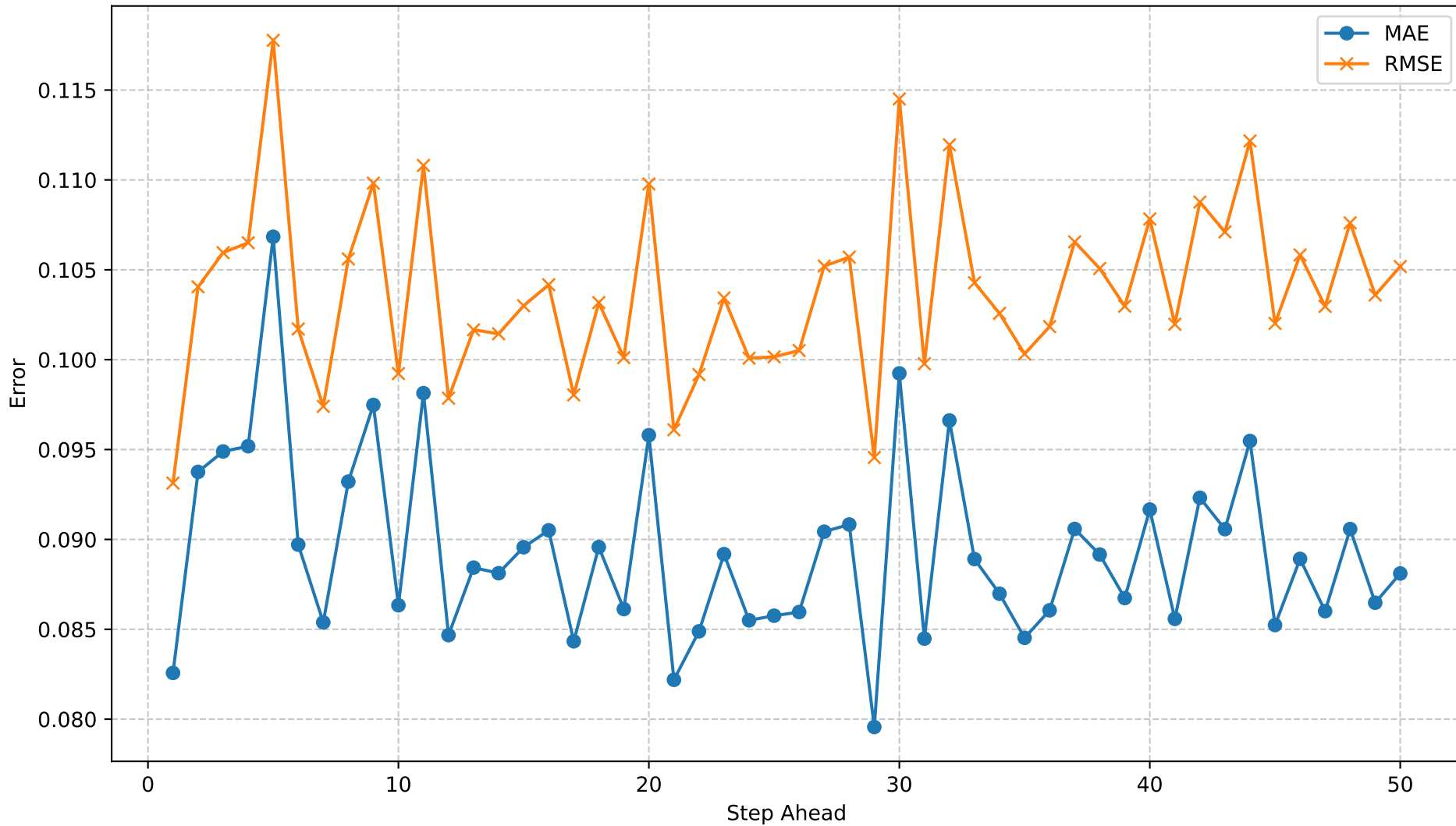
Predicted RSSI in Spatial Context (First Step)



Predicted RSSI in Spatial Context (Multi-Step)



Multi-Step Forecast Error by Step



Resumo Automático:

- Forecast Steps configurados: 50
- Passos com MAE ≤ 2.0 : 50 passo(s)
- MAE no último passo: 0.0881

[V] O modelo mantém boa acurácia durante quase toda a sequência de previsões.